

MACHINE LEARNING

Practice Question Paper – 50 Questions

Numericals + Theoretical Questions

Instructions:

All numerical datasets contain at most 5 samples. Show clear steps. Calculators are permitted. Answer all questions.

Total Questions: 50 | Numericals: 28 | Theoretical: 22

Section A – Numerical Questions

Q1. Linear Regression – OLS Closed Form (Matrix)

Given the data:

x	y
5	8
2	3.8
3	5.9
4	6.1

Using the closed-form solution $\beta = (X^T X)^{-1} X^T Y$, find the intercept and slope. (You may use a calculator for the inverse.) Model: $y = \beta_0 + \beta_1 x$

Q2. Linear Regression – Prediction & Residual

The fitted line is $y = 1.5x + 0.8$. For the point ($x = 4$, actual $y = 7.2$), calculate: (a) predicted y , (b) residual.

Q3. Equivalence of Logit and Sigmoid Forms in Logistic Regression

In logistic regression, the model is often written in two different forms:

Form 1 (Logit / Log-odds form):

$$\ln(p / (1 - p)) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

Form 2 (Sigmoid / Probability form):

$$P(Y = 1 | x) = 1 / (1 + e^{\{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)\}})$$

(a) Starting from Form 1, derive Form 2 step by step. Show every algebraic step clearly.

(b) Starting from Form 2, derive Form 1 step by step. Show every algebraic step clearly.

(c) What is the value of the log-odds when $P(Y = 1 | x) = 0.5$? What does this tell us about the decision boundary?

Q4. Logistic Regression – Probability

A logistic regression model is given by: $\ln(p / (1 - p)) = 1.2x - 0.6$. Calculate the probability p when $x = 1$.

Q5. Logistic Regression – Decision Boundary

Using the same model $\ln(p / (1 - p)) = 1.2x - 0.6$, find the value of x at which $p = 0.5$.

Q6. Confusion Matrix – Metrics

A classifier produced the following results:

	Predicted Positive	Predicted Negative
Actual Positive	3 (TP)	1 (FN)
Actual Negative	0 (FP)	1 (TN)

Calculate Accuracy, Precision, Recall and F1-score.

Q7. Confusion Matrix – Another Set

Given: TP = 2, FP = 1, FN = 1, TN = 1. Compute Precision, Recall, Accuracy and F1-score.

Q8. Information Gain – Select Best Feature

Dataset (5 samples) for binary classification (Play = Yes/No):

Outlook	Humidity	Wind	Play
Sunny	High	Weak	No
Sunny	High	Strong	No
Overcast	High	Weak	Yes
Rain	Normal	Weak	Yes
Rain	Normal	Strong	No

Calculate Information Gain for the feature “Outlook” and state whether it is a good split (show entropy calculations).

Q9. Information Gain – Best Feature Selection

Using the same 5-sample dataset as Q8, compute Information Gain for “Humidity” and “Wind”. Which feature should be selected as the root node? Justify.

Q10. KNN Classification

Training points (4 samples):

Class A: (1, 2), (2, 3) Class B: (5, 5), (6, 4)

Test point: (3, 3). Using K = 3 and Euclidean distance and Manhattan distance, predict the class. Show all distance calculations.

Q11. KNN Regression

Given points: (1, 10), (2, 15), (4, 25), (5, 30). Predict the value at $x = 3$ using K = 2 (Euclidean distance and Manhattan distance). Show steps.

Q12. Ensemble – Bagging (Majority Vote)

Three base classifiers trained on bootstrap samples give the following predictions on a test instance: Model-1 → Class A, Model-2 → Class B, Model-3 → Class A. Using majority voting (Bagging), what is the final prediction? Explain briefly why bagging reduces variance.

Q13. Ensemble – Boosting Concept

In a boosting iteration, three weak learners have weighted errors: $\epsilon_1 = 0.2$, $\epsilon_2 = 0.4$, $\epsilon_3 = 0.1$. Which learner receives the highest weight for the next round? Briefly explain the core idea of boosting.

Q14. Cross-Validation – Leave-One-Out

Dataset of 4 samples with true labels: [+ , + , - , -]. A model's leave-one-out predictions are: [+ , - , - , -]. Calculate the LOOCV accuracy.

Q15. Cross-Validation – 2-Fold

A dataset has 4 samples. In 2-fold cross-validation, Fold-1 accuracy = 75% and Fold-2 accuracy = 50%. What is the overall 2-fold CV accuracy?

Q16. Linear Regression – OLS Matrix Form

Data (4 samples):

x	y
10	10
11	30
12	40
13	60

Fit a simple linear regression model $y = w_0 + w_1x$, using the ordinary least squares closed-form solution.

Q17. Linear Regression – Multiple Features (2 predictors)

Data (3 samples):

x_1	x_2	y
1	2	5
2	1	4
3	3	8

Using OLS closed form, set up the matrices X and Y (include intercept column). Compute $X^T X$ and $X^T Y$ only (do not invert).

Q18. Logistic Regression – Logit to Probability

Given: $\ln(p/(1-p)) = -0.5 + 1.5x$. Calculate p when $x = 0$ and when $x = 2$. Also state the predicted class for each (threshold = 0.5).

Q19. Logistic Regression – Equivalence of Forms

Start from $\ln(p/(1-p)) = \theta_0 + \theta_1 x$ and derive the sigmoid form $p = 1 / (1 + e^{\{-(\theta_0 + \theta_1 x)\}})$. Show every algebraic step.

Q20. Confusion Matrix – Metrics from Counts

On a test set of 5 samples a model produced: TP=2, FP=1, FN=1, TN=1.

Calculate: (i) Accuracy (ii) Precision (iii) Recall (iv) F1-score. Also write the confusion matrix.

Q21. Confusion Matrix – From Predictions

True labels: [1, 0, 1, 1, 0] Predicted labels: [1, 0, 0, 1, 1]

Build the confusion matrix and compute Precision, Recall and F1-score for the positive class (1).

Q22. Information Gain – Entropy First

A dataset of 4 samples has target values: Yes, Yes, No, No.

Calculate the entropy of the entire dataset. Then, after splitting on a feature that puts {Yes, Yes} in one branch and {No, No} in the other, calculate Information Gain.

Q23. Information Gain – Choose Best Attribute

Dataset (4 samples):

Color	Size	Label
Red	Small	Yes
Red	Large	No
Blue	Small	Yes
Blue	Large	Yes

Calculate Information Gain for both “Color” and “Size”. Which attribute is better for the root node?

Q24. KNN Classification (K=1 and K=3)

Training data (5 samples):

A: (1,1), (2,2) B: (4,5), (5,4), (6,6)

Test point: (3,3). Predict the class using (i) K=1 and (ii) K=3 with Euclidean distance. Show distances.

Q25. Distance-Weighted KNN

For the same data as Q24, use distance-weighted nearest neighbour with K=3. Weight of a neighbour = $1/\text{distance}$. Predict the class of (3,3). Show weighted votes.

Q26. Ensemble – Bagging Numerical

Five bootstrap models predict the following classes for one test sample: A, B, A, A, B. Using majority voting, what is the bagged prediction? What is the vote margin?

Q27. Ensemble – Simple Boosting Weight Update

A weak learner has error $\epsilon = 0.25$ on the current weighted training set. Calculate the weight α that AdaBoost would assign to this learner ($\alpha = \frac{1}{2} \ln((1-\epsilon)/\epsilon)$).

Q28. Cross-Validation – 5-sample LOOCV

Five samples have true labels [1,1,0,0,1]. Leave-one-out predictions are [1,0,0,0,1]. Calculate LOOCV accuracy and the number of errors.

Section B – Theoretical Questions

Q29. Support Vector Machines (SVM)

Explain the difference between hard-margin and soft-margin SVM. What is the role of support vectors? Why is the maximum-margin hyperplane preferred?

Q30. Learning Problems

Differentiate among Supervised Learning, Unsupervised Learning and Reinforcement Learning with one real-world example for each.

Q31. Designing a Learning System

List and briefly explain the main components/steps involved in designing a machine learning system (as per Mitchell's framework or standard ML pipeline).

Q32. Perspectives and Issues in Machine Learning

Discuss any four important issues/perspectives in Machine Learning (e.g., overfitting, bias-variance trade-off, data quality, scalability, interpretability, ethical concerns).

Q33. Concept Learning Task

What is a Concept Learning Task? Define the terms: Instance Space, Hypothesis Space, Target Concept, and Training Examples with a simple example (e.g., "Enjoy Sport").

Q34. Concept Learning as Search & Version Spaces

Explain how concept learning can be viewed as a search problem. Define Version Space. How do the S-boundary and G-boundary represent the version space?

Q35. Inductive Bias

What is Inductive Bias in machine learning? Why is inductive bias necessary? Give examples of inductive bias in Candidate-Elimination algorithm and in Decision Tree learning.

Q36. Distance-Weighted Nearest Neighbour

Explain Distance-Weighted Nearest Neighbour algorithm. How does it differ from standard KNN? Write the formula used for weighting the contribution of neighbours.

Q37. Locally Weighted Regression

What is Locally Weighted Regression (LWR)? How does it differ from ordinary linear regression? Briefly describe the role of the kernel/weighting function.

Q38. Decision Trees – Hypothesis Space Search

Describe how Decision Tree learning performs a search through the hypothesis space. What is the role of Information Gain in this search? Mention one limitation of the basic ID3 algorithm.

Q39. Inductive Bias in Decision Tree Learning

Explain the inductive bias of the ID3 / Decision Tree learning algorithm. How does the preference for shorter trees arise? What happens if we allow unrestricted growth of the tree?

Q40. Cross-Validation – Hold-out vs K-fold idea

A dataset has only 4 samples. Explain why Leave-One-Out CV is preferred over a single 50-50 train-test split in this case. Also compute the number of models trained in LOOCV.

Q41. SVM – Soft Margin and Slack Variables

Why do we introduce slack variables in soft-margin SVM? What does a large value of the hyper-parameter C mean for the resulting decision boundary?

Q42. Learning Problems – Well-Posed Learning Problem

According to Tom Mitchell, what three components define a well-posed learning problem? Illustrate with the example of a chess-playing program that improves by playing against itself.

Q43. Designing a Learning System – Performance Element

In the standard design of a learning system, distinguish between the Performance Element, the Critic, and the Generalizer. How do they interact?

Q44. Perspectives and Issues – Overfitting

Define overfitting. Describe two practical techniques that help reduce overfitting in decision trees and in neural networks / logistic regression.

Q45. Concept Learning – Find-S Algorithm

Explain the Find-S algorithm for concept learning. What is its major limitation compared to the Candidate-Elimination algorithm?

Q46. Version Spaces – Specific and General Boundaries

In the Candidate-Elimination algorithm, what do the sets S and G represent? How is a new positive example used to update S , and how is a new negative example used to update G ?

Q47. Inductive Bias – Candidate Elimination vs ID3

Compare the inductive bias of the Candidate-Elimination algorithm with the inductive bias of the ID3 decision-tree algorithm. Which one is stronger and why?

Q48. Distance-Weighted Nearest Neighbour – Formula

Write the prediction formula used by distance-weighted k-NN for regression. Why does giving higher weight to closer neighbours usually improve performance over uniform weighting?

Q49. Locally Weighted Regression – Kernel Role

In Locally Weighted Regression, what is the purpose of the kernel (weighting) function? Name one common kernel and state how the bandwidth parameter affects the fit.

Q50. Decision Trees – Occam's Razor & Hypothesis Space Search

(a) How does the inductive bias of decision-tree learning relate to Occam's Razor? What would happen if we preferred deeper trees instead of shorter ones?

(b) Decision-tree algorithms perform a greedy search through the hypothesis space. Explain why the search is called "greedy" and mention one consequence of this greediness (e.g., local optima or attribute-selection bias).

— End of Question Paper —

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